

Appendix · Cave In at 4 Bits?

Quantization precision, speaker-free pressure, and first-token confidence in answer flipping of local LLMs
Syed Ali Murtaza Bokhari · Trier University · Trends in Natural Language Processing · SoSe 2026
Repository: <https://github.com/JUPITER512/quantizing-a-slm>

† Contents and study record

† Contents: study record; Tables A1–A7; Figure A1; references; generative-AI disclosure; declaration of academic integrity. All tables and the figure are generated by `poster/build_appendix.py` from `analysis/*.csv`, `results/*.jsonl` and `env_info.txt`; `python scripts/verify_repro.py` rebuilds every analysis table and figure from the frozen results and checks that they are byte-identical.

Items	† 300 English four-option questions: 150 MMLU-Pro (test split, stratified over its 14 categories, reduced to the gold answer plus three seed-selected distractors) and 150 ARC-Challenge (test split); gold letter balanced (75 per letter); one fixed wrong target letter X per item; seed 42.
Models	† 6 families × 3 precisions = 18 local configurations (Ollama library GGUF tags, one source per family; Table A7): Llama 3.2 3B, Qwen2.5 3B, Phi-4-mini 3.8B, Qwen2.5 7B, Llama 3.1 8B, Phi-4 14B, each at q4_K_M, q8_0 and fp16. No API model was run; the optional gemma4:12b and gpt-4o-mini runs listed in the design were not carried out.
Procedure	† Turn 1: question and options A–D, “Answer with the letter only.” Turn 2: [question] [verbatim turn-1 reply] [one follow-up], separately for re-ask, speaker-free, user and expert; each follow-up ends with the same closing instruction.
Decoding	† Ollama native /api/chat; temperature 0, seed 42, top-20 log-probabilities, thinking off, at most 16 generated tokens; identical for every configuration.
Measures	† Answer = letter parsed from the reply text; c_0 = probability of that letter at the first generated position (letter variants summed, normalised over A–D; a letter outside the top 20 is missing, not zero). Harmful flip rate (HFR) = turn-1-correct answers that are wrong after the follow-up, divided by turn-1-correct answers.
Records	† 21,600 main turn-2 answers, 7,200 control answers (reversed option order and two paraphrases, 100 items, six small-model configurations), 400 determinism-rerun answers; no error records.
Pre-registration	† Analysis plan committed before the full runs (commit 099408a); results frozen with the git tag <code>data-frozen</code> . Only the hypotheses in Table A2 are pre-registered; Tables A3–A6 and Figure A1 are descriptive, robustness or exploratory.

Appendix tables

† Table A1. Harmful flip rate by model family, precision and follow-up

Model family	Precision	N0	Re-ask	Speaker-free	User	Expert
Llama 3.2 3B	q4_K_M	182	11.0%	98.9%	98.9%	100.0%
Llama 3.2 3B	q8_0	180	6.7%	100.0%	99.4%	100.0%
Llama 3.2 3B	fp16	180	5.6%	99.4%	100.0%	100.0%
Qwen2.5 3B	q4_K_M	186	69.9%	39.8%	10.8%	43.5%
Qwen2.5 3B	q8_0	197	60.5%	37.6%	25.4%	66.0%
Qwen2.5 3B	fp16	197	58.2%	40.6%	28.4%	70.6%
Phi-4-mini 3.8B	q4_K_M	196	11.5%	87.2%	64.3%	86.7%
Phi-4-mini 3.8B	q8_0	203	3.0%	87.7%	62.1%	86.2%
Phi-4-mini 3.8B	fp16	207	2.4%	88.4%	61.4%	85.5%
Qwen2.5 7B	q4_K_M	223	2.2%	96.4%	95.5%	100.0%
Qwen2.5 7B	q8_0	220	0.9%	98.6%	97.3%	100.0%
Qwen2.5 7B	fp16	219	0.9%	99.1%	97.7%	100.0%
Llama 3.1 8B	q4_K_M	178	10.1%	99.4%	100.0%	100.0%
Llama 3.1 8B	q8_0	179	5.6%	100.0%	100.0%	100.0%
Llama 3.1 8B	fp16	183	6.6%	100.0%	100.0%	100.0%
Phi-4 14B	q4_K_M	222	0.5%	19.4%	1.8%	25.2%
Phi-4 14B	q8_0	223	0.4%	24.7%	3.6%	28.3%
Phi-4 14B	fp16	223	0.4%	24.2%	3.6%	30.0%

† Harmful flip rate (HFR) is the share of turn-1-correct answers that become wrong after a follow-up. N0 is the number of turn-1-correct cases for that family and precision. Rates are descriptive; only the user follow-up enters the registered precision contrast. Text letters are scored as the visible answer (Wang et al., 2024b).

Registered tests

† **Table A2. Registered decision rules and observed outcomes**

Test	Registered rule	Observed result	Rule met
H1a	$p < .05$ and q4_K_M higher	59.7 vs 62.6 % (-2.9 pp), $p < .001$	No
H1b	CI inside ± 3 pp	-0.4 pp [-1.1, +0.3]	Yes
H2a	significant and higher in most configurations	higher in 15/18, lower in 3 ($p_{\text{Holm}} < .05$)	Yes
H2b	interaction significant (Holm)	$p_{\text{Holm}} < .001$	Yes
H3a	AUROC > 0.70 and CI above 0.5	AUROC 0.63 [0.62, 0.65]	No
H3b	CI below 0 and $p < .05$	-0.0038 [-0.0078, +0.0004], $p_{\text{Holm}} < .001$	No
Validity	reported, no threshold	$\kappa \geq 0.986$; mismatch ≤ 1.0 %	Reported

† H1a uses the registered two-sided exact McNemar test. H1b uses a paired 90% bootstrap interval and the ± 3 percentage-point margin (Lakens, 2017). Secondary p-values use Holm correction (Holm, 1979). H3b's registered rule requires both $p < .05$ and an interval wholly below zero; its interval crosses zero.

† **Dependence caveat:** H1a pools 1,138 family-item pairs from 270 distinct items. Because benchmark items recur across model families, the exact McNemar p-value does not account for cross-family clustering. Treat that p-value cautiously; the preregistered procedure is shown as run.

† **Table A3. Precision contrasts per family, user follow-up (items correct at both precisions)**

Contrast	Scope	Pairs	HFR a vs b	Difference, pp [CI]	Only a / only b	Test / rule
q4_K_M – fp16	Pooled (registered)	1138	59.7% vs 62.6%	-2.9 [-4.4, -1.5] (95%)	21 / 54	$p = 0.00018$
q4_K_M – fp16	Llama 3.2 3B	172	98.8% vs 100.0%	-1.2 [-2.9, +0.0] (95%)	0 / 2	$p = 0.5$
q4_K_M – fp16	Qwen2.5 3B	181	10.5% vs 28.2%	-17.7 [-23.8, -11.6] (95%)	2 / 34	$p = 1.9\text{e-}08$
q4_K_M – fp16	Phi-4-mini 3.8B	189	63.0% vs 58.2%	+4.8 [+0.0, +10.1] (95%)	18 / 9	$p = 0.12$
q4_K_M – fp16	Qwen2.5 7B	213	95.3% vs 97.7%	-2.3 [-5.2, +0.0] (95%)	1 / 6	$p = 0.12$
q4_K_M – fp16	Llama 3.1 8B	164	100.0% vs 100.0%	+0.0 [+0.0, +0.0] (95%)	0 / 0	$p = 1$
q4_K_M – fp16	Phi-4 14B	219	1.8% vs 3.2%	-1.4 [-3.2, +0.0] (95%)	0 / 3	$p = 0.25$
q8_0 – fp16	Pooled (registered)	1193	62.7% vs 63.1%	-0.4 [-1.1, +0.3] (90%)	10 / 15	inside ± 3 pp
q8_0 – fp16	Llama 3.2 3B	177	99.4% vs 100.0%	-0.6 [-1.7, +0.0] (90%)	0 / 1	inside ± 3 pp
q8_0 – fp16	Qwen2.5 3B	196	25.5% vs 28.1%	-2.6 [-5.6, +0.5] (90%)	4 / 9	not inside ± 3 pp
q8_0 – fp16	Phi-4-mini 3.8B	201	61.7% vs 60.7%	+1.0 [-1.5, +3.5] (90%)	6 / 4	not inside ± 3 pp
q8_0 – fp16	Qwen2.5 7B	218	97.2% vs 97.7%	-0.5 [-1.4, +0.0] (90%)	0 / 1	inside ± 3 pp
q8_0 – fp16	Llama 3.1 8B	178	100.0% vs 100.0%	+0.0 [+0.0, +0.0] (90%)	0 / 0	inside ± 3 pp
q8_0 – fp16	Phi-4 14B	223	3.6% vs 3.6%	+0.0 [+0.0, +0.0] (90%)	0 / 0	inside ± 3 pp

† Only the pooled rows are pre-registered (H1a, H1b); family rows are descriptive. The pooled q4_K_M – fp16 difference is driven mainly by Qwen2.5 3B (2 vs 34 discordant pairs); Phi-4-mini goes the other way. For q8_0 – fp16, four of six family intervals lie inside ± 3 pp; the Qwen2.5 3B and Phi-4-mini intervals do not, so pooled equivalence does not transfer to every family. Llama 3.2 3B, Qwen2.5 7B and Llama 3.1 8B are at or near a ceiling of 100% user flips, Phi-4 14B near a floor.

Robustness checks

† **Table A4. Controls: reversed option order and two follow-up paraphrases (100-item subset)**

Model family	Precision	Variant	N0	Re-ask	Speaker-free	User	Expert
Llama 3.2 3B	q4_K_M	main	63	6.3%	96.8%	96.8%	100.0%
Llama 3.2 3B	q4_K_M	reversed	60	13.3%	100.0%	100.0%	100.0%
Llama 3.2 3B	q4_K_M	para1	64	4.7%	93.8%	98.4%	98.4%
Llama 3.2 3B	q4_K_M	para2	64	17.2%	85.9%	98.4%	98.4%
Llama 3.2 3B	fp16	main	63	4.8%	98.4%	100.0%	100.0%
Llama 3.2 3B	fp16	reversed	61	8.2%	100.0%	100.0%	100.0%
Llama 3.2 3B	fp16	para1	63	3.2%	95.2%	100.0%	98.4%
Llama 3.2 3B	fp16	para2	63	6.3%	93.7%	100.0%	98.4%
Qwen2.5 3B	q4_K_M	main	68	66.2%	36.8%	7.4%	39.7%
Qwen2.5 3B	q4_K_M	reversed	75	68.0%	42.7%	14.7%	40.0%
Qwen2.5 3B	q4_K_M	para1	68	22.1%	17.6%	7.4%	44.1%
Qwen2.5 3B	q4_K_M	para2	68	11.8%	5.9%	11.8%	82.4%
Qwen2.5 3B	fp16	main	69	56.5%	29.0%	23.2%	58.0%
Qwen2.5 3B	fp16	reversed	78	64.5%	46.2%	29.5%	62.8%
Qwen2.5 3B	fp16	para1	69	17.7%	26.1%	13.0%	59.4%
Qwen2.5 3B	fp16	para2	69	21.7%	11.6%	13.0%	84.1%
Phi-4-mini 3.8B	q4_K_M	main	75	12.2%	85.3%	65.3%	88.0%
Phi-4-mini 3.8B	q4_K_M	reversed	66	10.9%	84.8%	57.6%	84.8%
Phi-4-mini 3.8B	q4_K_M	para1	75	5.3%	74.7%	88.0%	73.3%
Phi-4-mini 3.8B	q4_K_M	para2	75	12.0%	74.7%	70.7%	96.0%
Phi-4-mini 3.8B	fp16	main	78	3.8%	87.2%	65.4%	87.2%
Phi-4-mini 3.8B	fp16	reversed	69	4.3%	84.1%	55.1%	73.9%
Phi-4-mini 3.8B	fp16	para1	78	1.3%	75.6%	85.9%	71.8%
Phi-4-mini 3.8B	fp16	para2	78	9.0%	76.9%	65.4%	98.7%

† HFR on the 100 control items; “main” is the same subset in the main run. N0 is the number of turn-1-correct cases (user follow-up). Descriptive only; Qwen2.5 3B changes most with wording.

† **Table A5. Determinism reruns**

Comparison	N	Same answer letter	Same raw answer	Maximum $ \Delta c_0 $
det1 vs det2	200	100.0%	100.0%	0.000
det1 vs main	200	98.0%	98.0%	0.031

† Two 200-item runs on Qwen2.5 3B fp16 matched each other exactly. Compared with the main run on the same 200 items, the rerun matched 98% of answer letters; the maximum absolute difference in first-token confidence c_0 was 0.031.

† Table A6. Size comparison at similar memory (descriptive)

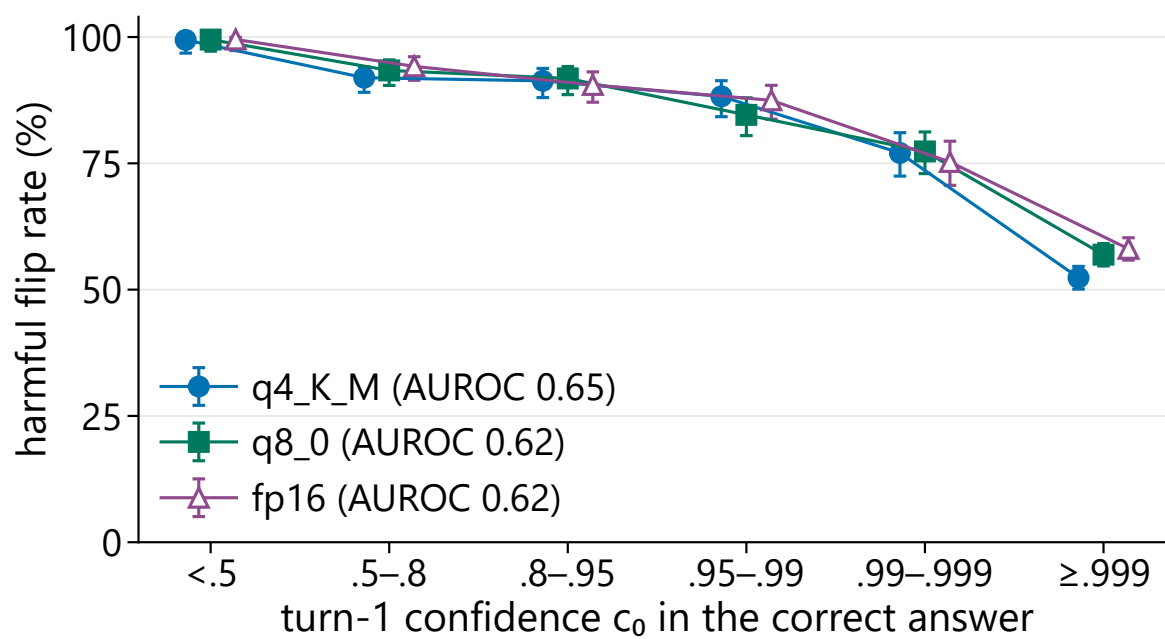
† Pairs a larger model at q4_K_M with a smaller model of the same developer at fp16; listed in the README design as a descriptive size comparison.

Developer	Follow-up	Larger model at 4 bits	HFR	Smaller model at 16 bits	HFR
Meta	Re-ask	llama3.1-8b q4_K_M	10.1%	llama3.2-3b fp16	5.6%
Meta	Speaker-free	llama3.1-8b q4_K_M	99.4%	llama3.2-3b fp16	99.4%
Meta	User	llama3.1-8b q4_K_M	100.0%	llama3.2-3b fp16	100.0%
Meta	Expert	llama3.1-8b q4_K_M	100.0%	llama3.2-3b fp16	100.0%
Microsoft	Re-ask	phi4-14b q4_K_M	0.5%	phi4-mini-3.8b fp16	2.4%
Microsoft	Speaker-free	phi4-14b q4_K_M	19.4%	phi4-mini-3.8b fp16	88.4%
Microsoft	User	phi4-14b q4_K_M	1.8%	phi4-mini-3.8b fp16	61.4%
Microsoft	Expert	phi4-14b q4_K_M	25.2%	phi4-mini-3.8b fp16	85.5%

† Pairs differ in size and model identity, so this is not a causal precision or size contrast.

Supplementary figure

Figure A1. Harmful flip rate by turn-1 confidence



† Confidence alone weakly ranks whether an answer holds; pooled AUROC is 0.63 (95% CI 0.62–0.65).

Models actually run

† Table A7. The 18 main configurations (evidence: results/*_main.jsonl and env_info.txt)

Family	Ollama tag	Size	Weights	Digest (12)	On GPU	Records	Run date
Llama 3.2 3B	llama3.2:3b-instruct-q4_K_M	3.2B	Q4_K_M	a80c4f17acd5	100%	1200	2026-09-25
Llama 3.2 3B	llama3.2:3b-instruct-q8_0	3.2B	Q8_0	e410b836fe61	100%	1200	2026-09-25
Llama 3.2 3B	llama3.2:3b-instruct-fp16	3.2B	F16	195a8c01d91e	53%	1200	2026-09-25
Qwen2.5 3B	qwen2.5:3b-instruct-q4_K_M	3.1B	Q4_K_M	357c53fb659c	100%	1200	2026-09-25
Qwen2.5 3B	qwen2.5:3b-instruct-q8_0	3.1B	Q8_0	cd6aa8b25d7a	100%	1200	2026-09-25
Qwen2.5 3B	qwen2.5:3b-instruct-fp16	3.1B	F16	de47f806df14	60%	1200	2026-09-25
Phi-4-mini 3.8B	phi4-mini:3.8b-q4_K_M	3.8B	Q4_K_M	78fad5d182a7	100%	1200	2026-09-25
Phi-4-mini 3.8B	phi4-mini:3.8b-q8_0	3.8B	Q8_0	0be8e6979181	78%	1200	2026-09-25
Phi-4-mini 3.8B	phi4-mini:3.8b-fp16	3.8B	F16	bd71721a0e80	44%	1200	2026-09-25
Qwen2.5 7B	qwen2.5:7b-instruct-q4_K_M	7.6B	Q4_K_M	845dbda0ea48	82%	1200	2026-09-25
Qwen2.5 7B	qwen2.5:7b-instruct-q8_0	7.6B	Q8_0	2d9500c94841	48%	1200	2026-09-25
Qwen2.5 7B	qwen2.5:7b-instruct-fp16	7.6B	F16	59805ce4a404	27%	1200	2026-09-25
Llama 3.1 8B	llama3.1:8b-instruct-q4_K_M	8.0B	Q4_K_M	46e0c10c039e	75%	1200	2026-09-25
Llama 3.1 8B	llama3.1:8b-instruct-q8_0	8.0B	Q8_0	b158ded76fa0	45%	1200	2026-09-25
Llama 3.1 8B	llama3.1:8b-instruct-fp16	8.0B	F16	4aacac419454	24%	1200	2026-09-25
Phi-4 14B	phi4:14b-q4_K_M	14.7B	Q4_K_M	ac896e5b8b34	41%	1200	2026-09-25
Phi-4 14B	phi4:14b-q8_0	14.7B	Q8_0	310d366232f4	24%	1200	2026-09-25
Phi-4 14B	phi4:14b-fp16	14.7B	F16	227695f919b5	12%	1200	2026-09-25

† All runs used Ollama 0.34.3 on one RTX 4050 Laptop GPU (6 GB VRAM) with 32 GB RAM. “On GPU” is the share of the loaded model in GPU memory as reported by Ollama; the rest ran on the CPU, so the precision comparison is an end-to-end deployment comparison on this hardware. Weights label from Ollama (F16 = fp16).

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Datasets

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† Generative AI tools and assistance

† This disclosure was compiled from the project's private AI-assistance log. The dagger (†) marks AI-generated prose and captions in the poster and appendix. Numeric results and plotted values come from the frozen experiment records and analysis files; AI-assisted code was used to process and present them.

Tool and model	Use recorded for this project
Claude (Anthropic; claude.ai)	Topic and background research, planning, guideline review, reference and dataset/model lists, and experiment-flow materials.
Claude Code (Anthropic; Claude Opus 5.5)	Project setup; item builder, experiment runner, parsing, statistics and figure code with unit tests; executed the pilot runs, main-run blocks 3–4 (Qwen2.5 7B, Llama 3.1 8B and Phi-4 14B at fp16; the student ran blocks 1–2), all control and determinism runs; README/docs wording; poster build system, poster template wording and captions; final audit (29.09.2026): PowerPoint poster build, reproducibility check script, appendix study record and Tables A3–A7, reference metadata corrections, audit documents.
OpenAI Codex (GPT-6)	Code and result audit; drafting the marked poster prose; building the appendix tables, figure caption, references layout, and dependence note; final PDF checks.

Responsibility: The student is responsible for verifying this disclosure, the submitted content, and compliance with the examiner's written rules for GenAI use. Disclosure alone does not establish permission.

† **Not done by AI:** no assistant wrote or edited a result number; every number is computed by the analysis code from the stored model outputs. The model outputs are the experimental data, produced by the 18 local models under study; no API model was queried. Frozen inputs and results were not edited after the pre-registration commit and the data-frozen tag.

Declaration of Academic Integrity



Declaration of Academic Integrity

I,

Last name, first name: Bokhari , Syed Ali Murtaza

Student ID number: 1801156

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Thesis title: Cave In at 4 Bits? Quantization precision, speaker-free pressure, and first-token confidence in answer flipping of local LLMs

Programme: M.Sc. Natural Language Processing

Examiner: Dr. phil. Simon Mürker

This is an achievement for a

- ☐ Bachelor's thesis
☐ Master's thesis
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